INFEED

Simulated product
Presence feedback

TEST NEST

Position / clamp
PLC request ID

VIRTUAL ATE

Simulated voltage
PASS / FAIL / RETEST

SORT / OUTFLOW

Gate feedback
Completion count

PLC-01 / LS XBC-DR32H

XG-SIM: MAIN sequence, output commands, alarms.
SIM_IO: plant response and safety input model.
Environment gates new-product START only.

PLC-02 / FX3U-class

GX Simulator: environmental judgment.
Separate prepare / commit / ACK sensor interface.
Exact physical model suffix still to confirm.

SENSOR-01 / ESP32-S3

SEN0137 real temperature and humidity.
Wi-Fi Modbus TCP sensor server.
No motor control / ATE role.

SCADA-01 / CIMON 3.91

Operator pages, alarms, inspection display.
Two controlled bridges: ESP → FX and FX → LS.
LS driver, OPC client and virtual-ATE server endpoints.

ATE-01 / LabVIEW VirtualATE

Modbus TCP client to CIMON; simulated mV test.
Records PLC-accepted inspection in CSV.
No real DUT electrical testing or wafer-map integration.

PASS=1, FAIL=2, RETEST=3. RETEST is a separate result bin; automatic re-test loop is not implemented.

Physical conveyor, clamps, gates and safety protection have not been commissioned. See drawings 018-022 for the planned design.

| NO. | DATE | REVISION |
|---|---|---|
| R0 | 2026-10-22 | FUNCTION RECORD |

DWG TITLE

SEMICON SORTER PROJECT
Final Test and Multi-Bin Sorting Cell

SYSTEM ROLES AND PROCESS SCOPE

| DRAWN BY | DATE |
|---|---|
| J. H. KO | 2026-10-13 |

DWG NO

002

SHEET NO

1 OF 1